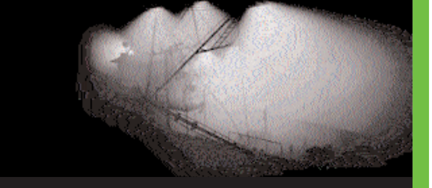
create a realistic view inside Gollum's skin. The filmmakers also brought in character actor Andy Serkis to give Gollum a range of voices—from melancholy to menacing.

Brewing the Perfect Storm

George Lucas and his special effects company, Industrial Light and Magic (ILM), have a special place of mention in the world of movie special effects. ILM began in the 1970s with the cult series *Star Wars* (1977). ILM also digitally created the realistic-looking dinosaurs for *Jurassic Park*. John Anderson, in-house scientist for ILM, used fluid dynamics to develop a simulation of the ocean that could be manipulated for *The Perfect Storm*. The simulation needed to handle everything from rippling, still water, to waves the size of a 10-storey building. Basic fluid dynamic simulation provided the foundation for the body of the ocean, and became known as the bottom water in the film.

Most of the actual 3D modelling work (*see box, 'Bringing a 3D Model to Life'*) was done in Maya, made by Alias Wavefront which uses C++ to allow animators and designers to write their own custom plug-ins. *The Perfect Storm* team wrote more than 30 plug-ins for Maya for this movie. They also wrote several stand-alone applications for specific aspects, such as shaders and particle systems of the ocean scenes.

Splash created by boat crashing into water



Boat's lighting model

Frame of the final shot with all elements integrated

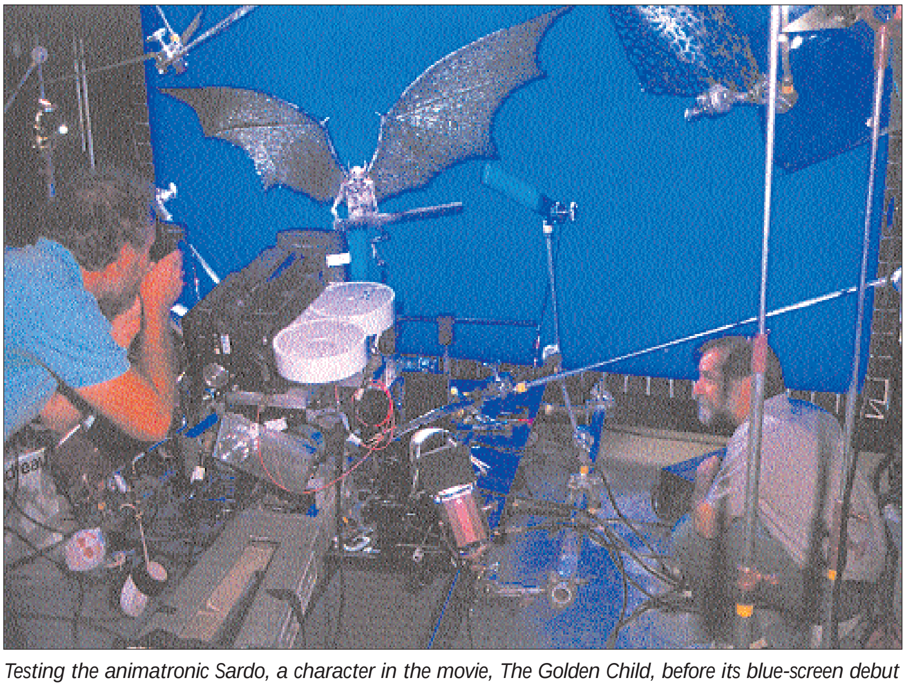
Photo Courtesy: ILM

Post-production digital magic

A typical film might have 1,000 to 1,500 shots in it, and they get spliced together one after the other to create the whole film. The film then enters the post-production phase. It is here that a major part of the computer magic potion is introduced.

Now, the parts of the film in which digital effects need to be added are digitised. Special Effects need a holistic understanding of how the audio-visual sensory parts of our body and brain perceive the world around us, and then use this information to simulate events. It's important because the Principle of Motion Picture is based on the phenomenon of persistence of vision—our eyes can retain a picture for a fraction of a second. If a series of still pictures is flashed before the eyes in rapid succession, the eyes see it as a scene depicting smooth action.

Compositing, a special effects technique, relies on the way our mind perceives motion. The object and its reference points are necessary to perceive motion in a scene. Compositing is the process of adding all of the different layers of a shot together to create the final shot. The compositing artist incorporates all the elements in the right order so that they overlay each other properly to create the final shot. Directors now extensively use a motion-control camera which is controlled by a computer, and can be programmed to precisely duplicate the same movement repeatedly. With motion control, multiple elements can be filmed in exactly the same way, allowing the images to be aligned for compositing.



Testing the animatronic Sardo, a character in the movie, The Golden Child, before its blue-screen debut

In this context, blue and green screens (technically called Chromakeying) may have intrigued you. Do you recall the cycles flying in the film *ET*? In *Star Wars*, Luke flies his X-wing fighter down the trench of the Death Star with Twin Ion Engine fighters in close pursuit. The X-wing fighter, TIE fighters and the trench were all models. All these illusions were created by a special effects technique known as travelling matte or blue screen. The models are filmed separately against blue backgrounds and the footage is then combined in multiple layers to make the final film. The blue screen lets you combine two or more pieces of film into one. The actor cannot wear anything blue, as it will show up as a hole. With computers, blue screen shots are easier because computers can create the mattes and combine the shots automatically.

Animatronics

A related technology and much older technique is Animatronics. Animatronic creatures are quite life-like—think *Jaws* and *Jurassic Park*. An animatronic is a mechanised puppet that may be pre-programmed or remotely controlled. A complicated animatronic could take up to two years from conception to completion.